ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** infinitely many solutions (coincident lines)	$\frac{a_1}{a_2}=\frac{1}{2}$, $\frac{b_1}{b_2}=\frac{1}{2}$, $\frac{c_1}{c_2}=\frac{1}{2}$
2	**(a)** $(3, 2)$	$2(3)+3(2)=12=12$
3	**(a)** 7	$k=2(2)+3(1)$
4	**(c)** 7	$x+y=10$, $x-y=4$ gives $2x=14$
5	**(a)** y-axis	All points have the same x-coordinate
6	**(b)** coincident	One equation is a multiple of the other
7	**(b)** no solution (parallel lines)	$\frac{a_1}{a_2}=\frac{1}{2}$, $\frac{b_1}{b_2}=\frac{1}{2}$, $\frac{c_1}{c_2}=\frac{5}{7}$
8	**(d)** $(2, 5)$	$3(2)+1(5)=11=11$
9	**(a)** 14	$k=3(4)+1(2)$
10	**(a)** 10	$x+y=14$, $x-y=6$ gives $2x=20$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{a_1}{a_2}=\frac{1}{2}$, $\frac{b_1}{b_2}=\frac{1}{2}$ (1); all ratios equal, infinitely many solutions, consistent (1).	1+1
12	Adding: $2x=16$ (1) so $x=8$, $y=4$ (1).	1+1
13	$k=3(2)+2(5)$ (1) $=16$ (1).	1+1
14	$\frac{a_1}{a_2}=\frac{1}{2}$, $\frac{b_1}{b_2}=\frac{1}{2}$ (1); $a_1/a_2=b_1/b_2\neq c_1/c_2$, no solution, inconsistent (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Multiply (i) by -1 and (ii) by 1 and subtract: $(-2)x=-10$ (2); $x=5$ (1); put in (i): $y=3$ (1); check in (ii) (1).	2+1+1+1
16	From (i): $y=\frac{18-2x}{3}$ (1); substitute in (ii) (1); simplify to $(1)x=3$ (1); $x=3$ (1); $y=4$ (1).	1+1+1+1+1
17	Let numbers be x, y: $x+y=25$, $x-y=11$ (1); adding $2x=36$ (1); numbers 18 and 7 (1).	1+1+1
	OR: Multiply (i) by -1 and (ii) by 1 and subtract: $(-3)x=-12$ (2); $x=4$ (1); put in (i): $y=2$ (1); check in (ii) (1).	2+1+1+1
18	From (i): $y=\frac{12-x}{3}$ (1); substitute in (ii) (1); simplify to $(5)x=30$ (1); $x=6$ (1); $y=2$ (1).	1+1+1+1+1
	OR: Let numbers be x, y: $x+y=34$, $x-y=16$ (1); adding $2x=50$ (1); numbers 25 and 9 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Multiply (i) by 4 and (ii) by 3 and subtract: $(-7)x=-21$ (2); $x=3$ (1); put in (i): $y=2$ (1); check in (ii) (1).	2+1+1+1
	OR: Let pencil $=x$, eraser $=y$: $2x+3y=22$, $3x+2y=23$ (2); solving, $x=5$, $y=4$ (2); pencil ₹5, eraser ₹4 (1).	2+2+1
20	Multiply (i) by -3 and (ii) by 2 and subtract: $(-13)x=-52$ (2); $x=4$ (1); put in (i): $y=1$ (1); check in (ii) (1).	2+1+1+1
	OR: Let pencil $=x$, eraser $=y$: $4x+y=27$, $x+3y=15$ (2); solving, $x=6$, $y=3$ (2); pencil ₹6, eraser ₹3 (1).	2+2+1

Mistakes to avoid:

- Applying the ratio test with the wrong pairing, e.g. comparing $\frac{a_1}{b_2}$ instead of $\frac{a_1}{a_2}$.
- Saying a pair with $\frac{a_1}{a_2}=\frac{b_1}{b_2}$ always has no solution; check $\frac{c_1}{c_2}$ to separate coincident lines from parallel lines.
- Subtracting only some terms in elimination, or forgetting to multiply the right-hand side as well as the left.
- Reading the result $0=0$ as 'no solution'; it signals infinitely many solutions, while $0=5$ signals none.
- Not substituting back to find the second variable, or stating only x when the question asks for both.
- Writing the reversed number as $10x+y$ again instead of $10y+x$ in digit problems.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** y-axis	All points have the same x-coordinate
2	**(c)** coincident	One equation is a multiple of the other
3	**(d)** exactly one solution (intersecting lines)	$\frac{a_1}{a_2}=\frac{1}{2}$, $\frac{b_1}{b_2}=\frac{3}{5}$, $\frac{c_1}{c_2}=\frac{4}{9}$
4	**(c)** $(3,3)$	$1(3)+2(3)=9=9$
5	**(a)** 13	$k=1(3)+5(2)$
6	**(a)** 14	$x+y=20$, $x-y=8$ gives $2x=28$
7	**(b)** y-axis	All points have the same x-coordinate
8	**(b)** coincident	One equation is a multiple of the other
9	**(a)** infinitely many solutions (coincident lines)	$\frac{a_1}{a_2}=\frac{1}{2}$, $\frac{b_1}{b_2}=\frac{1}{2}$, $\frac{c_1}{c_2}=\frac{1}{2}$
10	**(d)** $(2,3)$	$2(2)+1(3)=7=7$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Adding: $2x=26$ (1) so $x=13$, $y=7$ (1).	1+1
12	$k=4(3)-1(2)$ (1) $=10$ (1).	1+1
13	$\frac{a_1}{a_2}=3$, $\frac{b_1}{b_2}=-2$ (1); ratios differ, one solution, consistent (1).	1+1
14	Adding: $2x=18$ (1) so $x=9$, $y=6$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Multiply (i) by -1 and (ii) by 2 and subtract: $(-7)x=-21$ (2); $x=3$ (1); put in (i): $y=1$ (1); check in (ii) (1).	2+1+1+1
16	From (i): $y=\frac{11-3x}{1}$ (1); substitute in (ii) (1); simplify to $(-2)x=-4$ (1); $x=2$ (1); $y=5$ (1).	1+1+1+1+1
17	Let numbers be x,y: $x+y=43$, $x-y=21$ (1); adding $2x=64$ (1); numbers 32 and 11 (1).	1+1+1
	OR: Multiply (i) by 1 and (ii) by 1 and subtract: $(2)x=4$ (2); $x=2$ (1); put in (i): $y=5$ (1); check in (ii) (1).	2+1+1+1
18	From (i): $y=\frac{5-x}{2}$ (1); substitute in (ii) (1); simplify to $(7)x=21$ (1); $x=3$ (1); $y=1$ (1).	1+1+1+1+1
	OR: Let numbers be x,y: $x+y=53$, $x-y=27$ (1); adding $2x=80$ (1); numbers 40 and 13 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Multiply (i) by 2 and (ii) by 3 and subtract: $(-1)x=-3$ (2); $x=3$ (1); put in (i): $y=2$ (1); check in (ii) (1).	2+1+1+1
	OR: Let pencil $=x$, eraser $=y$: $3x+2y=34$, $5x+y=45$ (2); solving, $x=8$, $y=5$ (2); pencil ₹8, eraser ₹5 (1).	2+2+1
20	Multiply (i) by 4 and (ii) by 2 and subtract: $(14)x=28$ (2); $x=2$ (1); put in (i): $y=5$ (1); check in (ii) (1).	2+1+1+1
	OR: Let pencil $=x$, eraser $=y$: $2x+5y=24$, $6x+y=44$ (2); solving, $x=7$, $y=2$ (2); pencil ₹7, eraser ₹2 (1).	2+2+1

Mistakes to avoid:

- Applying the ratio test with the wrong pairing, e.g. comparing $\frac{a_1}{b_2}$ instead of $\frac{a_1}{a_2}$.
- Saying a pair with $\frac{a_1}{a_2}=\frac{b_1}{b_2}$ always has no solution; check $\frac{c_1}{c_2}$ to separate coincident lines from parallel lines.
- Subtracting only some terms in elimination, or forgetting to multiply the right-hand side as well as the left.
- Reading the result $0=0$ as 'no solution'; it signals infinitely many solutions, while $0=5$ signals none.
- Not substituting back to find the second variable, or stating only x when the question asks for both.
- Writing the reversed number as $10x+y$ again instead of $10y+x$ in digit problems.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 10	$k = 4(1) + 2(3)$
2	**(b)** 10	$x + y = 18$, $x - y = 2$ gives $2x = 20$
3	**(c)** y-axis	All points have the same x-coordinate
4	**(b)** coincident	One equation is a multiple of the other
5	**(a)** no solution (parallel lines)	$\frac{a_1}{a_2} = \frac{1}{3}$, $\frac{b_1}{b_2} = \frac{1}{3}$, $\frac{c_1}{c_2} = \frac{1}{2}$
6	**(c)** $(2, 4)$	$5(2) + 1(4) = 14 = 14$
7	**(a)** 16	$k = 5(2) + 3(2)$
8	**(c)** 21	$x + y = 30$, $x - y = 12$ gives $2x = 42$
9	**(d)** exactly one solution (intersecting lines)	$\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = -1$, $\frac{c_1}{c_2} = \frac{3}{5}$
10	**(a)** $(2, 1)$	$4(2) + 1(1) = 9 = 9$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$k = 2(1) + 7(1)$ (1) $= 9$ (1).	1+1
12	$\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$ (1); all ratios equal, infinitely many solutions, consistent (1).	1+1
13	Adding: $2x = 40$ (1) so $x = 20$, $y = 10$ (1).	1+1
14	$k = 5(2) + 3(-1)$ (1) $= 7$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Multiply (i) by 1 and (ii) by 3 and subtract: $(-5)x = -30$ (2); $x = 6$ (1); put in (i): $y = 2$ (1); check in (ii) (1).	2+1+1+1
16	From (i): $y = \frac{10 - 2x}{1}$ (1); substitute in (ii) (1); simplify to $(3)x = 12$ (1); $x = 4$ (1); $y = 2$ (1).	1+1+1+1+1
17	Let numbers be x, y: $x + y = 41$, $x - y = 13$ (1); adding $2x = 54$ (1); numbers 27 and 14 (1).	1+1+1
	OR: Multiply (i) by 1 and (ii) by 3 and subtract: $(-1)x = -3$ (2); $x = 3$ (1); put in (i): $y = 4$ (1); check in (ii) (1).	2+1+1+1
18	From (i): $y = \frac{8 - x}{1}$ (1); substitute in (ii) (1); simplify to $(2)x = 10$ (1); $x = 5$ (1); $y = 3$ (1).	1+1+1+1+1
	OR: Let numbers be x, y: $x + y = 53$, $x - y = 37$ (1); adding $2x = 90$ (1); numbers 45 and 8 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Multiply (i) by 2 and (ii) by -3 and subtract: $(13)x = 78$ (2); $x = 6$ (1); put in (i): $y = 2$ (1); check in (ii) (1).	2+1+1+1
	OR: Let pencil $= x$, eraser $= y$: $5x + 2y = 38$, $2x + 3y = 35$ (2); solving, $x = 4$, $y = 9$ (2); pencil ₹4, eraser ₹9 (1).	2+2+1
20	Multiply (i) by -2 and (ii) by 4 and subtract: $(-26)x = -104$ (2); $x = 4$ (1); put in (i): $y = 2$ (1); check in (ii) (1).	2+1+1+1
	OR: Let pencil $= x$, eraser $= y$: $3x + 4y = 51$, $4x + y = 42$ (2); solving, $x = 9$, $y = 6$ (2); pencil ₹9, eraser ₹6 (1).	2+2+1

Mistakes to avoid:

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- Saying a pair with $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ always has no solution; check $\frac{c_1}{c_2}$ to separate coincident lines from parallel lines.
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